

- 2 Lionfish are a type of fish native to the tropical waters of the Indian Ocean and Pacific Ocean.

Lionfish were first reported off the south east coast of the USA in the 1980s. Lionfish are an alien species to this area. Their lack of a natural predator in these waters resulted in their numbers increasing dramatically, allowing them to migrate to new areas. They have now become established across the Caribbean. Their numbers are so high that they are now a threat to many native species.

- (a) In one year it was estimated that 5 million prey were consumed by 1000 lionfish.

Express these data as a ratio.

..... : 1 [1]

- (b) Surveys to assess the changes in lionfish abundance have been carried out every year since 2004. Divers have carried out visual transects at nine sites along a 15km stretch of continuous reef off the coast of one island in the Caribbean.

Visual transects involve swimming in a fixed direction across the reef for a set distance and counting the number of lionfish.

- (i) State **one** variable which was standardised in these surveys.

.....
..... [1]

The mean abundance of lionfish on the reef was calculated for each year. These values are shown in Fig. 2.1 using a log scale. The bars show 95% confidence intervals.

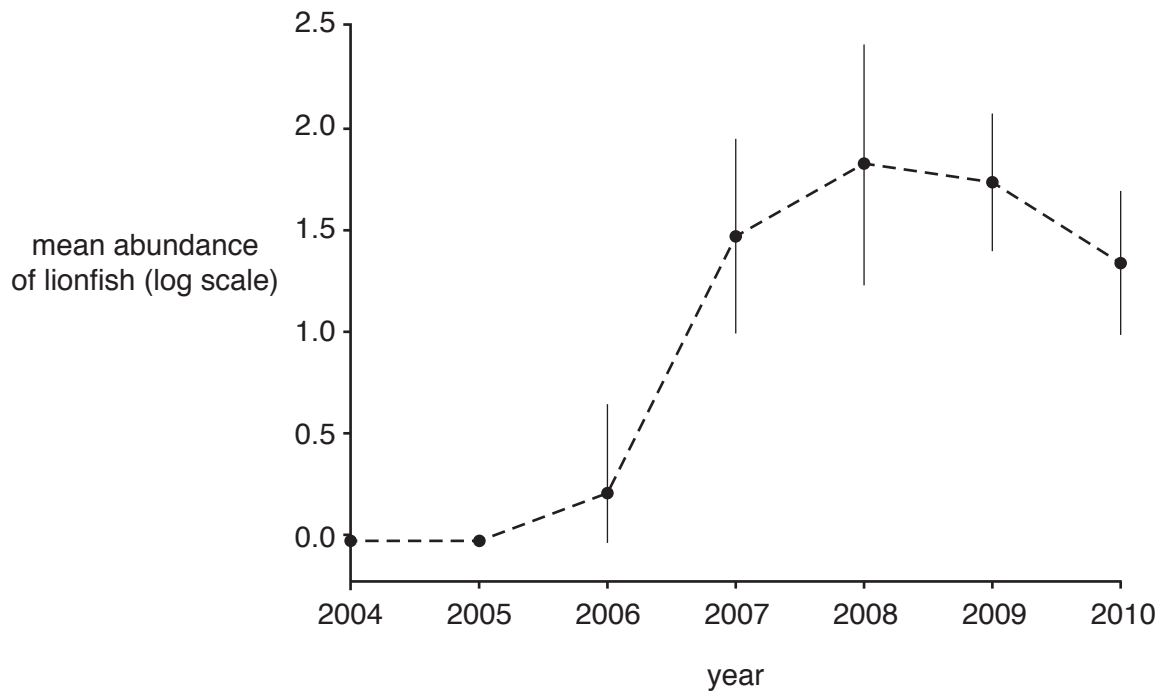


Fig. 2.1

- (ii) A group of students read an article stating that local people were being encouraged to catch and eat lionfish in an attempt to reduce their numbers.

The students concluded that local people started catching and eating lionfish in 2008.

Explain to what extent the data in Fig. 2.1 supports their conclusion.

.....

.....

.....

.....

.....

..... [3]

- (c) Some local communities hold annual fishing trips where they are encouraged to catch as many lionfish as possible.

In 2010 and 2011, 16 boats took part in these fishing trips. The mean number of lionfish per boat and the 95% confidence intervals were calculated.

Fig. 2.2 shows the results.

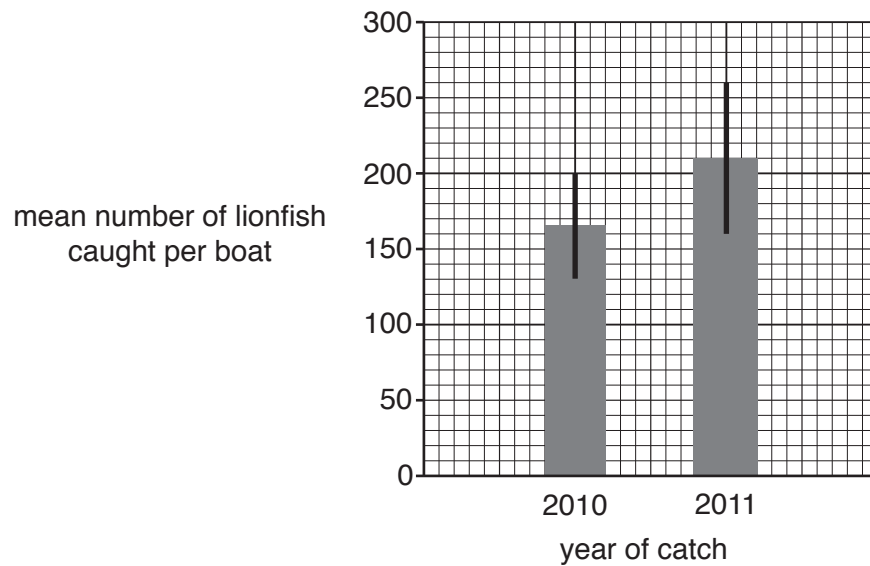


Fig. 2.2

The 95% confidence intervals are calculated from $2 \times S_M$.

Standard deviation can be calculated by rearranging the equation for standard error shown in Fig. 2.3.

$$S_M = \frac{s}{\sqrt{n}}$$

where S_M = standard error

s = standard deviation

n = number of boats

Fig. 2.3

Use the data in Fig. 2.2 and the equation in Fig. 2.3 to calculate the standard deviation of the mean number of lionfish caught per boat in 2010.

standard deviation for 2010 = [3]

[Total: 8]